

SYSTEMATIC REVIEW

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# Effectiveness of school-based interventions on fundamental movement skills in children: a systematic review and meta-analysis

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## Abstract

**Background** Fundamental movement skills (FMS) are essential prerequisites for children's active participation in physical activities (PA), which plays a crucial role in promoting both physical and mental health, ultimately contributing to their overall well-being. Children spend a significant portion of their time in school, making it a critical setting for the development of their FMS. However, systematic reviews and meta-analyses of the effects of school-based interventions on FMS have not been summarized. Therefore, the aim of this study was to analyze the effectiveness of school-based interventions on FMS in children.

**Methods** This study employed a systematic review and meta-analysis of studies published between 2014 and 2024. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed. A total of 15,930 publications were searched in the Web of Science (WOS), PubMed, and EBSCOhost databases. Two rounds of literature screening were conducted, including duplicate removal and title and abstract screening. Data analysis was conducted using Review Manager version 5.4, with meta-analysis performed using a random effects model. Publication bias was assessed using Stata version 18. The quality of the included studies was evaluated using the Physiotherapy Evidence Database (PEDro) scale.

**Results** A total of thirty-three studies from 14 countries were included in the analysis. Among them, thirty (90.91%) studies demonstrated that their interventions were effective. Thirty-one (93.94%) studies were implemented within classroom settings. Twenty-one (63.64%) studies had an intervention duration of less than 60 min. Twelve (36.36%) studies were conducted twice a week. Sixteen (48.48%) studies lasted less than 10 weeks. Meta-analysis of nine interventions indicated significant effects on overall FMS proficiency (SMD = 0.69, 95% CI = 0.21–1.16,  $I^2 = 94\%$ ). Meta-analysis indicated that interventions with a duration of 60 min or more, a frequency of 3 times or more a week, and a period of 10 to 20 weeks were efficacious at enhancing FMS in children.

**Conclusions** School-based interventions are effective in promoting children's FMS. Long-duration, high-frequency, medium- and long-period interventions may be optimal for enhancing FMS.

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**Keywords** Fundamental movement skills, Intervention, Children, Systematic review, Meta-analysis, Effectiveness, School

## Background

Fundamental movement skills (FMS) are essential movement patterns that don't naturally develop and serve as the foundation for more complex physical activities (PA). FMS can be categorized into three main types: locomotor skills (e.g., walking, running, jumping), object control skills (e.g., throwing, catching), and stability skills (e.g., balancing) [1]. These skills are considered primary predictors of children's motor competence and are critical for the future acquisition of a diverse range of movement sequences. Proficiency in FMS plays a significant role in enhancing children's PA, fosters self-perception, and supports executive function [2]. The development of FMS is closely linked to cardiorespiratory endurance and other health-related components [3–5]. A deficiency in FMS can impede the development of specific movement abilities in children, which can affect their participation in PA and can lead to various health-related issues, such as obesity. Current literature highlights the importance of FMS as a fundamental element in children's overall growth and development. However, it's concerning that the proficiency of FMS among children has declined in many countries [6, 7].

The acquisition of FMS is influenced by various factors in different environments. Factors such as the availability of exercise equipment and space within the school, family, and community play a significant role in the development of FMS [8]. It's essential that teachers, parents, and other community members receive quality instruction, hands-on opportunities, feedback, and knowledge of FMS and education, as these factors can potentially influence the development of FMS [9]. Previous studies have summarized the effects of PA interventions, active play (AP) interventions, and some other interventions on FMS in different environments, demonstrating various interventions in different settings can promote the development of children's FMS [10, 11].

No institution or environment has a greater influence on an individual during the first two decades of their life than the school [12]. On average, children spend a substantial portion of their time in school, approximately half of the year [13]. Furthermore, schools provide numerous opportunities for participation in various activities throughout the weekdays, including physical education (PE) classes and recess [14, 15]. Teacher training, education materials, and curriculum design are all crucial to promoting FMS, PA, and health-related content in school [12]. Moreover, the World Health Organization (WHO) has identified schools as critical environments for enhancing PA among children,

advocating for curriculum-based interventions that address the entire student body to maximize intervention reach, while recommending that schools offer appropriate planning, opportunities, and diverse venues for PA [16]. Consequently, some countries, such as Poland [17] and the United States [18], are evaluating the relationship between the proficiency of FMS and PE classes. Furthermore, in China, FMS is listed as part of the compulsory education standards [19]. The significance of the school environment in the learning, acquisition, and reinforcement of FMS is further emphasized because FMS also plays a solid foundation for intentions and attitudes to be physically active in adulthood [20].

Most interventions focus on schools, with PE instructors acting as the primary facilitators. These interventions aim to enhance children's FMS through diverse programs, including badminton [21] taekwondo [22], and some others. Despite the variety in the intervention content, the inconsistency in parameters such as duration, frequency, and period of interventions makes it difficult to determine the optimal approach. One study proposed that the duration of the intervention may not be a crucial factor in determining its impact [23]. A review identified two interventions, one lasting 4 weeks and the other 36 weeks, with no significant differences between the results of two studies [24]. Conversely, another review, suggested that interventions for the enhancement of children's FMS should last at least 60 min in duration, 3 times per week or more, and last for a minimum of 20 weeks [25]. Additionally, there is a pressing need to enhance the diversity of assessment tools, along with the reliability and validity of their outcomes [26]. A further synthesis of evidence regarding interventions on FMS in school settings is needed.

In 2013, a systematic review and meta-analysis summarized interventions in schools, families, and communities, noting that the majority of these interventions were implemented within school settings [27]. However, the study did not separately analyze the interventions carried out in the school setting. Consequently, this review aimed to summarize the effectiveness of school-based interventions on FMS among children aged 6 to 12 years (a critical period for FMS development) with a focus on studies conducted in the past decade (2014–2024). Additionally, we aimed to offer evidence-based recommendations and implications for interventions designed to improve children's FMS within the school setting. This study also examined the duration, frequency, and period across different studies.

## Methods

This systematic review and meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [28]. The study was registered in the International Prospective Register for Systematic Review (PROSPERO).

### Search strategy

This study aims to analyze the effectiveness of school-based interventions on children's FMS. The following databases were searched from 2014 to 2024: Web of Science (WOS) (Web of Science Core Collection, KCI-Korean Journal Database, Preprint Citation Index, ProQuest™ Dissertations & Theses Citation Index, Scielo Citation Index), PubMed (MEDLINE, life science journals, online books), and EBSCOhost (Academic Search Premier, ERIC, MEDLINE, MEDLINE Complete, SPORTDiscus with Full Text, Regional Business News, Business Source Premier, Humanities International Complete, eBook Collection, Newspaper Source, International Bibliography of Theatre & Dance with Full Text, Film & Television Literature Index with Full Text). The key search terms were divided into four groups: (i) FMS: *fundamental movement skill or fundamental motor skill or motor competence or gross motor skill or motor skill or motor development or movement skill or coordination or locomotive skill or object control skill or ball skill or stability skill*; (ii) population: *children or kid or youth or teen\* or child\* or student\* or primary or elementary*; (iii) intervention: *intervention\* or training or method or approach or training programme\* or pilot or random\* or clinic\* or trial or experiment or feasibility or exercise*; (iv) environment: *school*.

### Inclusion and exclusion criteria

The inclusion criteria for screening articles were as follows: (i) the target population is healthy children aged 6 to 12 years; (ii) the study type involves randomized controlled trial (RCT), quasi-experimental study, including control group and intervention group, pre-test and post-test; (iii) the interventions in school settings; (iv) the outcomes include FMS; (v) full text published in English.

The exclusion criteria for screening articles were as follows: (i) the population of infants, adults, and children with disorders, obesity or various diseases; (ii) non-intervention study (e.g. reviews, cross-sectional study); (iii) the interventions were not conducted in schools; (iv) outcomes not include FMS indicators; (v) publication language with non-English.

### Study selection and data extraction

All articles from selected databases were imported into Endnote 20 and checked for duplicates. Two investigators reviewed the titles and abstracts of the remaining

articles. Articles agreed for inclusion were reviewed in full text. The difference between reviewers was managed through discussion, and disagreement was handled by a third party.

Data extraction was completed separately by two authors, and inconsistent information was judged by the third author based on the articles. Information on the first author, year, country, age, study type, intervention contents, assessment tools, and effects were contained in the data extraction.

### Quality assessment

The quality of the included studies was assessed using the Physiotherapy Evidence Database (PEDro) scale, which consists of four parts with 11 items. To ensure the effectiveness of the study quality assessment, two authors independently completed the quality assessment of each study design according to the PEDro scale, and the third author assessed the results of any disagreement between the two authors. The assessment scores were YES (1 point) and NO (0 point), and the scores of each study were summed up as the evaluation score of study quality.

### Statistical methods and data analysis

The meta-analysis included studies that utilized the Test of Gross Motor Development (TGMD) for assessment and included pre-tests and post-tests for both the intervention and control groups. In the process of meta-analysis, to improve the comparability of results and reduce the impact of different assessment tools on the results. Therefore, we only included the TGMD as an assessment tool. Meta-analysis was performed using Review Manager version 5.4, and the standardized mean difference (SMD) and 95% confidence interval (95% CI) were used as the effect scales to calculate the combined effect values. Heterogeneity of the included literature was tested using  $I^2$ , when  $P < 0.05$  and  $I^2 > 50\%$ , heterogeneity was large, and a random effects model was used, otherwise a fixed effects model was used. In this study, publication bias was tested by Stata version 18 using Egger's linear regression. To determine if the heterogeneity between studies was due to specific studies, the overall effect was analyzed by examining each study individually.

## Results

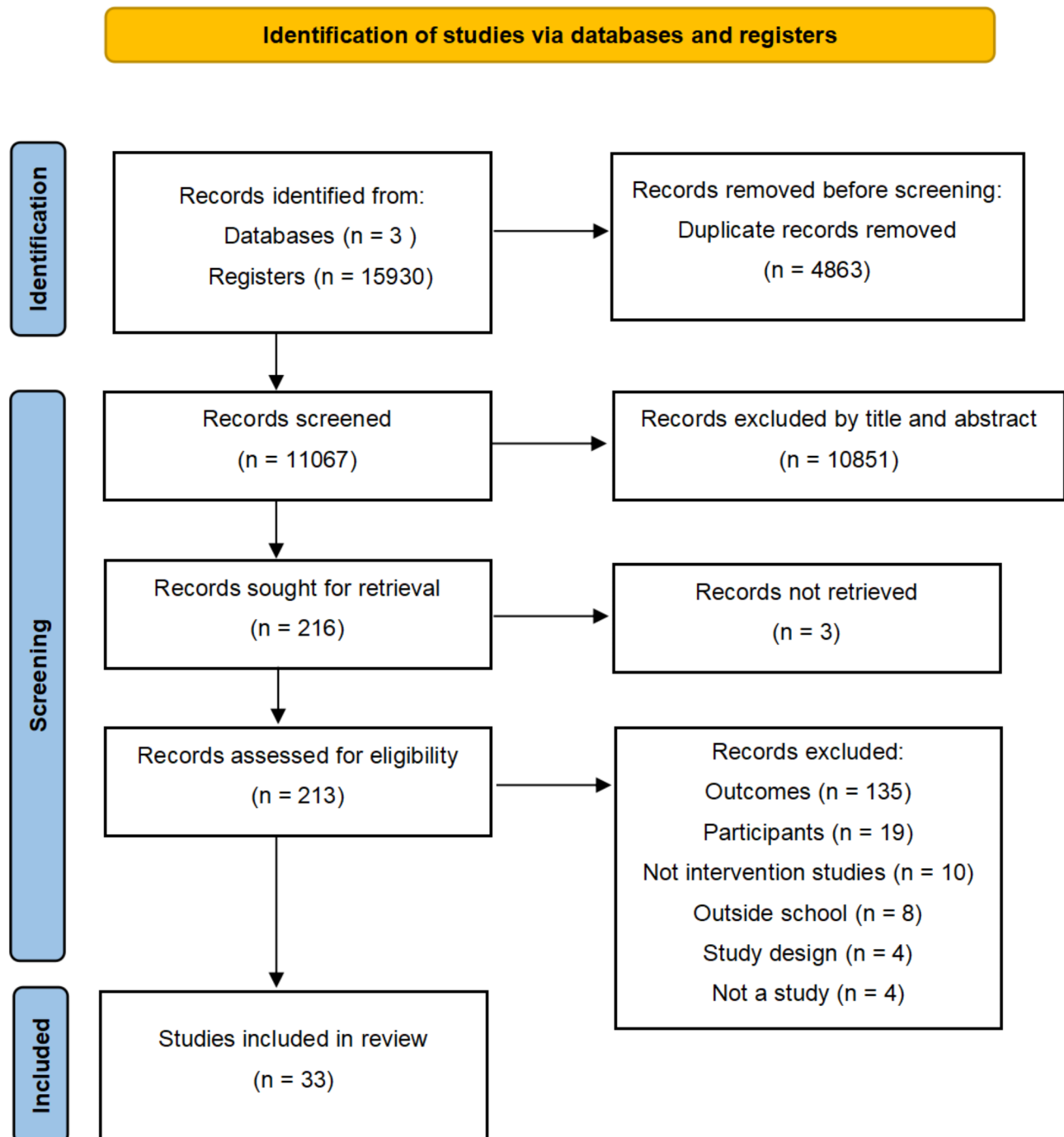
### Study selection

The search yielded a total of 15,930 English language articles, of which 5733 were found in WOS, 4878 in PubMed, and 5319 in EBSCOhost. The retrieved literature was imported into Endnote 20, and after removing duplicates, there were a total of 11,067 articles. After screening the articles by title and abstract, 216 studies were included. During the screening of the full papers by two authors, 27 papers had differing opinions from the 213 full papers,

and a consistent discussion was conducted. Afterward, two papers still showed differences and were reviewed by a third author. Finally, after screening the full texts, 33 studies [21, 22, 26, 29–58] were included. The process of screening the literature is shown in Fig. 1.

### Characteristics of included studies

The characteristics of the literature are shown in Table 1. The thirty-three literatures with 6571 participants in this study from 14 countries were included from 2014 to June 2024, with the majority from the US (18.18%) [31, 33, 35, 36, 48, 57], UK (15.15%) [21, 43, 52, 54, 55], Ireland (9.09%) [30, 37, 53], Australia (9.09%) [29, 39, 40], Indonesia (9.09%) [34, 41, 51], Spain (6.06%) [22, 38], Scotland



**Fig. 1** PRISMA flowchart for study selection

**Table 1** Study characteristics

| Study          | Country  | Age         | Study Type                | Intervention Characteristics |                                                                                 |                  | Contents                                                                                                                                       | Assessment   | Effect                                                        |
|----------------|----------|-------------|---------------------------|------------------------------|---------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------|
|                |          |             |                           | Dura-<br>tion<br>(min)       | Frequency                                                                       | Period<br>(week) |                                                                                                                                                |              |                                                               |
| Culjak 2014    | Croatia  | 7 ± 0.5     | Quasi-experimental design | 45                           | 3                                                                               | 18               | Basic artistic gymnastics program                                                                                                              | FMS -polygon | POST > PRE                                                    |
| Burns 2017a    | US       | 8.4 ± 1.8   | Quasi-experimental design | 50                           | 1                                                                               | 12               | Comprehensive School Physical Activity Program                                                                                                 | TGMD-2       | POST > PRE                                                    |
| Burns 2017b    | US       | 9.1 ± 1.5   | Quasi-experimental design | 50                           | 1                                                                               | 32               | Comprehensive School Physical Activity Program                                                                                                 | TGMD-3       | POST > PRE                                                    |
| Johnstone 2017 | Scotland | 7.0 ± 1.1   | Quasi-experimental design | 60                           | 1                                                                               | 20               | Go2Play Active Play, structured games, and free play                                                                                           | TGMD-2       | INT > CON                                                     |
| Elif 2018      | Spain    | 7–10        | RCT                       | 60                           | 3                                                                               | 12               | Taekwondo training                                                                                                                             | BOT-2        | INT > CON                                                     |
| Duncan 2018    | UK       | 6–7         | RCT                       | 30–40                        | 1                                                                               | 10               | Structure of the 10-week integrated neuromuscular training (INT) program                                                                       | TGMD-2       | INT > CON                                                     |
| Bolger 2019    | Ireland  | 6 and 10    | Quasi-experimental design | Girls: 25;<br>Boys: 30       | 2                                                                               | 26               | Intervention 1: physical activity<br>Intervention 2: multicomponent fundamental movement skill                                                 | TGMD-2       | Intervention 1: 0<br>Intervention 2: INT > CON                |
| Johnstone 2019 | Scotland | 7           | RCT                       | 60                           | 1                                                                               | 10               | Active play intervention: 1 h outdoor physical activity session: 30 min of facilitated games; 30 min of free play                              | TGMD-2       | INT > CON                                                     |
| Zhang 2019     | China    | 6–9         | Quasi-experimental design | 40                           | 2                                                                               | 12               | PE-based program. (a) 5-min warm-up games, (b) 30-min of LOG tasks (FMS instruction and practice), and (c) 5-min cool down.                    | TGMD-2       | INT > CON                                                     |
| Chan 2019      | China    | 8.4 ± 0.56  | RCT                       | 45                           | 1                                                                               | 13               | An assessment-based intervention that emphasizes fun, mastery, and support (A + FMS)                                                           | TGMD-3       | INT > CON                                                     |
| Rodriguez 2019 | Spain    | 6–12        | Quasi-experimental design | 90                           | 1                                                                               | 8                | Balance intervention programs a direct instructional model (DIM) or a tactical games model (TGM).                                              | MABC-2       | INT > CON                                                     |
| Donahoe 2019   | US       | 10–12       | Quasi-experimental design | 40                           | Weeks1,5: once a week<br>Weeks2,3,4,8: twice a week<br>Weeks6,7: 3 times a week | 8                | Yoga sessions                                                                                                                                  | BOT-2        | POST > PRE: Balance; Flexibility 0: coordination and strength |
| Duncan 2020    | UK       | 8.5 ± 1.    | RCT                       | 50                           | 1                                                                               | 6                | Shuttle Time intervention                                                                                                                      | TGMD-2       | INT > CON                                                     |
| Lee 2020a      | US       | 6.65 ± 0.98 | Quasi-experimental design | 60                           | 3                                                                               | 8                | 60 min:10 min of instruction and preparation, 45 min of skill instruction and practice, and 5 min in a closing activity                        | TGMD-2       | INT > CON                                                     |
| Lee 2020b      | US       | 6.52 ± 0.97 | Quasi-experimental design | 60                           | 3                                                                               | 8                | (a) 10 min of warming-up preparation for FMS lessons and activities, (b) 45 min of focused FMS practices, and (c) 5 min of a closing activity. | TGMD-2       | INT > CON                                                     |

| Study            | Country   | Age                                  | Study Type                | Intervention Characteristics |                                                                                                                                                                |                  | Contents                                                                                                                                                                                          | Assessment                        | Effect     |
|------------------|-----------|--------------------------------------|---------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|------------|
|                  |           |                                      |                           | Dura-<br>tion<br>(min)       | Frequency                                                                                                                                                      | Period<br>(week) |                                                                                                                                                                                                   |                                   |            |
| Oppici 2020      | Australia | 8.8±0.7                              | RCT                       | 60                           | 2                                                                                                                                                              | 7                | Dance PE curriculum with different teaching pedagogies and levels of cognitive challenge                                                                                                          | CAMSA                             | INT > CON  |
| Costello 2020    | Ireland   | 9±1                                  | Quasi-experimental design | 30                           | 2                                                                                                                                                              | 4                | Movement literacy Intervention                                                                                                                                                                    | Fundamental Motor Skills Quotient | INT > CON  |
| Kelly 2020       | Ireland   | Year 3 (6.9±0.4)<br>Year 4 (7.9±0.4) | RCT                       | 45                           | 2                                                                                                                                                              | 8                | Intervention sessions were delivered using the principles of the TARGET acronym (i.e. task, authority, recognition, grouping, evaluation, and time) to facilitate a mastery-motivational climate. | TGMD-3                            | INT > CON  |
| Grainger 2020    | UK        | 10–11                                | RCT                       | 50–60                        | 2                                                                                                                                                              | 4                | Integrated FMS with a strength training intervention                                                                                                                                              | CAMSA                             | INT > CON  |
| Pham 2021        | Vietnam   | 7                                    | Quasi-experimental design | 35                           | 2                                                                                                                                                              | 20               | PE integrated with BRAINballs                                                                                                                                                                     | TGMD-2                            | INT > CON  |
| Moura 2021       | Brazil    | 8.00±0.86                            | Quasi-experimental design | 50                           | 1                                                                                                                                                              | 12               | Learn-to-swim programs                                                                                                                                                                            | KTK                               | POST > PRE |
| Rudd 2021        | Australia | 6.6±0.5                              | RCT                       | 50                           | 2                                                                                                                                                              | 8                | Dance curriculums                                                                                                                                                                                 | CAMSA                             | 0          |
| Duncan 2021      | UK        | 7.6±0.97                             | RCT                       | 40                           | 1 or 2                                                                                                                                                         | 10               | Badminton World Federation (BWF) Shuttle Time program                                                                                                                                             | TGMD-2                            | INT > CON  |
| Magistro 2022    | Italy     | 6.6±0.3                              | Quasi-experimental design | 60                           | T1–2: 8 h class per week(1 h normal PE and 7 h intervention activity in math class )<br>T3–4: 7 h class per week(1 h normal PE and 6 h intervention activity ) | 2 years          | Integrated physical activity in all mathematics teaching hours                                                                                                                                    | TGMD-3                            | INT > CON  |
| Rizkyanto 2023   | Indonesia | Grade 4                              | Quasi-experimental design | 45                           | 2                                                                                                                                                              | 12               | Brain Jogging Model                                                                                                                                                                               | TGMD-3                            | INT > CON  |
| Pratt 2023       | UK        | 7.8±0.63                             | RCT                       | 30–45                        | 1                                                                                                                                                              | 6                | Aquatic motor competence program                                                                                                                                                                  | TGMD-3                            | INT > CON  |
| Costa 2024       | US        | 9.1                                  | Quasi-experimental design | 120                          | 1                                                                                                                                                              | 12               | Super Quinas                                                                                                                                                                                      | MCA                               | INT > CON  |
| Nicolosi 2024    | Italy     | 8.4±0.3                              | Quasi-experimental design | 60                           | 2                                                                                                                                                              | 12               | PE lessons for 1 h per week, and interdisciplinary PE lessons for 1 h per week                                                                                                                    | TGMD                              | INT > CON  |
| Mardiansyah 2024 | Indonesia | 9                                    | Quasi-experimental design | 45                           | 2                                                                                                                                                              | 8                | Game-Based Motor Coordination (GBMC) learning model                                                                                                                                               | KTK3+                             | INT > CON  |

| Study          | Country      | Age         | Study Type                | Intervention Characteristics |                      |                  | Contents                                                                                | Assessment | Effect     |
|----------------|--------------|-------------|---------------------------|------------------------------|----------------------|------------------|-----------------------------------------------------------------------------------------|------------|------------|
|                |              |             |                           | Dura-<br>tion<br>(min)       | Frequency            | Period<br>(week) |                                                                                         |            |            |
| Cheong 2024    | Pakistan     | 8.17 ± 1.34 | Quasi-experimental design | /                            | 18 sessions in total | 6                | Chindro random, Chindro game-based, Pittu random, and Pittu game-based                  | TGMD-2     | POST > PRE |
| Idamokoro 2024 | South Africa | 7.12 ± 0.71 | Quasi-experimental design | 30                           | 2                    | 9                | warm-up exercises; aerobic fitness; ball skills; locomotor exercises; balance exercises | TGMD-3     | INT > CON  |
| Bikalawan 2024 | Indonesia    | 9–10        | Quasi-experimental design | /                            | 4 sessions in total  | 4                | flash card learning media                                                               | TGMD-2     | POST > PRE |
| Sortwell 2024  | Australia    | 8–10        | RCT                       | 7–10                         | each school day      | 6                | plyometric-based structured game active breaks                                          | CAMSA      | INT > CON  |

INT: Intervention Group; CON: Control Group; TGMD-2: Test of Gross Motor Development, Second Edition; TGMD-3: Test of Gross Motor Development, Third Edition; CAMSA: Canadian Agility and Movement Skill Assessment; BOT-2: Bruininks-Oseretsky Test of Motor Proficiency-Second Edition; FMS-polygon: Fundamental Movement Skill Polygon; KTK: Körperkoordinationstest für Kinder; MCA: Motor Competence Assessment; FMSQ: Fundamental Motor Skills Quotient; MABC-2: Movement Assessment Battery for Children-Second Edition

(6.06%) [26, 47], China (6.06%) [44, 50], Italy (6.06%) [56, 58], Vietnam (3.03%) [32], Croatia (3.03%) [42], Brazil (3.03%) [49], Pakistan (3.03%) [46], and South Africa (3.03%) [45].

### Descriptions of included studies

Most studies suggested their interventions were effective. Thirty (90.91%) studies [21, 22, 26, 30–52, 54–56, 58] showed that the interventions were effective. One (3.03%) study [29] showed that intervention was not effective. One (3.03%) study [57] indicated that the intervention was effective in balance and flexibility but not in coordination and strength. One (3.03%) study [53] showed that the results of PA intervention were not effective and the intervention of multi-component FMS was effective.

Interventions are mostly implemented through PA in classes and breaks. Interventions in thirty-one (93.94%) included studies [21, 22, 26, 29–38, 40, 42–58] that were implemented through different sports in classes. One (3.03%) study [39] employed plyometric-based structured game active breaks to promote children's FMS. One (3.03%) study [41] through non-PA, offered intervention in flash card learning media.

The duration of the interventions in the included studies ranged from 25 to 120 min. There were twenty-one (63.64%) studies [21, 29, 30, 32–34, 37, 39, 42–45, 48–55, 57] with an intervention duration of less than 60 min. Ten (30.30%) studies [22, 26, 31, 35, 36, 38, 40, 47, 56, 58] offered 60 min or more. And two (6.06%) studies [41, 46] didn't specify the duration.

Twelve (36.36%) studies [29, 30, 32, 34, 37, 40, 43–45, 51, 53, 56] offered twice a week. Eleven (33.33%) studies [26, 31, 33, 38, 47–50, 52, 54, 55] provided interventions once a week. Five (15.15%) studies [22, 35, 36, 42, 58] employed 3 times a week or more. One (3.03%) study [57]

used different frequencies, with the frequency of once a week in weeks 1 and 5, twice a week in weeks 2, 3, 4, and 8, and 3 times a week in weeks 6 and 7. One (3.03%) study [21] employed a frequency of once or twice a week. Three (9.09%) studies [39, 41, 46] didn't specify the frequency.

The intervention period ranged from 4 weeks to 2 academic years. Sixteen (48.48%) studies [29, 30, 34–41, 43, 45, 46, 52, 55, 57] lasted less than 10 weeks. Fourteen (42.42%) studies [21, 22, 26, 31–33, 42, 44, 47, 49–51, 54, 56] lasted 10 to 20 weeks. And three (9.09%) studies [48, 53, 58] lasted more than 20 weeks.

### Descriptions of assessment tools

Twenty-one (63.64%) studies [21, 26, 32, 33, 35–37, 41, 44–48, 50–56, 58] used the Test of Gross Motor Development (TGMD) instrument to assess children's FMS. Four (12.12%) studies [29, 39, 40, 43] used the Canadian Agility and Movement Skill Assessment (CAMSA) to assess children's FMS. Two (6.06%) studies [34, 49] used Körperkoordinationstest für Kinder (KTK) as an assessment tool. Two (6.06%) studies [22, 57] used the Bruininks-Oseretsky Test of Motor Proficiency-Second Edition (BOT-2). One (3.03%) study [31] used the Motor Competence Assessment (MCA). One (3.03%) study [30] used the Fundamental Motor Skills Quotient. One (3.03%) study [42] used a Fundamental Movement Skill Polygon (FMS-polygon). One (3.03%) study [38] used a Movement Assessment Battery for Children-Second Edition (MABC-2).

### Quality assessment

The quality of the included studies in this review was assessed independently by two co-authors using the PEDro scale. The Cohen's Kappa coefficient between the two raters is 0.704 (95% CI = 0.522–0.887,  $P < 0.001$ ),

Table 2 Quality assessment

| First author, Year | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9 | T10 | T11 | Scores |
|--------------------|----|----|----|----|----|----|----|----|----|-----|-----|--------|
| Culjak, 2014       | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0   | 1   | 2      |
| Burns, 2017a       | 1  | 1  | 1  | 0  | 0  | 0  | 0  | 0  | 1  | 0   | 1   | 4      |
| Burns, 2017b       | 1  | 1  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Johnstone, 2017    | 1  | 1  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 1   | 1   | 5      |
| Elif, 2018         | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 1   | 1   | 4      |
| Duncan, 2018       | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 1   | 1   | 7      |
| Bolger, 2019       | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Johnstone, 2019    | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 8      |
| Zhang, 2019        | 1  | 1  | 0  | 1  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 7      |
| Donahoe, 2019      | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0   | 1   | 2      |
| Rodriguez, 2019    | 1  | 1  | 1  | 0  | 0  | 0  | 0  | 0  | 1  | 1   | 1   | 5      |
| Chan, 2019         | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 8      |
| Duncan, 2020       | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 1   | 1   | 7      |
| Lee, 2020a         | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Lee, 2020b         | 1  | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 6      |
| Oppici, 2020       | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 7      |
| Costello, 2020     | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Kelly, 2020        | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 8      |
| Grainger, 2020     | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 0  | 1  | 1   | 1   | 6      |
| Pham, 2021         | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 1   | 1   | 4      |
| Moura, 2021        | 1  | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 1   | 1   | 7      |
| Rudd, 2021         | 1  | 1  | 0  | 0  | 1  | 1  | 1  | 1  | 1  | 1   | 1   | 8      |
| Duncan, 2021       | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 7      |
| Magistro, 2022     | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 7      |
| Rizkyanto, 2023    | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Pratt, 2023        | 1  | 1  | 1  | 0  | 1  | 0  | 0  | 0  | 1  | 1   | 1   | 6      |
| Costa, 2024        | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Nicolosi, 2024     | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 1   | 1   | 5      |
| Mardiansyah, 2024  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 4      |
| Cheong, 2024       | 0  | 1  | 1  | 0  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 6      |
| Bikalawan, 2024    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 0   | 1   | 3      |
| Idamokoro, 2024    | 1  | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 1  | 1   | 1   | 6      |
| Sortwell, 2024     | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 1  | 1  | 1   | 1   | 7      |

T1 Eligibility criteria specified; T2 Random allocation; T3 Concealed allocation; T4 Groups similar at baseline; T5 Participant blinding; T6 Therapist blinding; T7 Assessor blinding; T8 Fewer than 15% dropouts; T9 Intention-to-treat analysis; T10 Between-group statistical comparisons; T11 Point measures and variability data

indicating a high level of agreement and reliability in the quality assessment conducted by the raters. The results of the literature quality assessment were discussed after the discussion of the inconsistent results, and the results of the quality assessment are shown in Table 2. Scores on the PEDro scale ranged from 0 to 10, with scores < 4 being poor, 4 to 5 being fair, 6 to 8 being good, and 9 to 10 being excellent. In the included studies, scores ranged from 2 to 8. A total of three (9.09%) [41, 42, 57] of the included studies scored < 4, thirteen (39.39%) [22, 30–35, 38, 47, 48, 51, 53, 56] scored 4 to 5 and seventeen (51.52%) studies [21, 26, 29, 36, 37, 39, 40, 43–46, 49, 50, 52, 54, 55, 58] scored 6 to 8.

### Meta-analysis

In the process of meta-analysis, to improve the comparability of results and reduce the impact of different assessment tools on the results. The following 3 conditions should be met: (i) the study included the intervention group and the control group; (ii) the study reported the pre-test data of the intervention group and the control group; (iii) the assessment tool was TGMD. Thus, nine studies were included in meta-analysis. In the included intervention studies, after data (mean, standard deviation (SD) or standard error (SE)) extraction, the pre-test and post-test data of the reported intervention and control groups will be calculated and the obtained change values will be analyzed, where the following formula was used:

$$SD = SE * \sqrt{n}$$

$$\Delta Mean = Mean_{post-test} - Mean_{pre-test}$$

$$\Delta SD = \sqrt{[(SD_{pre-test})^2 + (SD_{post-test})^2 - 2 * R * SD_{pre-test} * SD_{post-test}]}$$

$R = 0.5$ ;  $n$ : Sample Size

### Overall effects of included interventions on FMS

A total of nine interventions in included studies with 1360 participants were included in the meta-analysis. The results showed a moderate effect but there was considerable heterogeneity (SMD = 0.69, 95% CI = 0.21–1.16,  $I^2 = 94\%$ ; Fig. 2). The results were analyzed in subgroups

according to the different intervention duration, frequencies, and periods.

### Effect of intervention duration on FMS

Figure 3a presented interventions with a duration of less than 60 min (SMD = 0.50, 95% CI = -0.19–1.19,  $I^2 = 96\%$ ) displayed no significant effect on FMS. Interventions with a duration of 60 min or more (SMD = 0.93, 95% CI = 0.28–1.58,  $I^2 = 84\%$ ) displayed a significant effect on FMS.

### Effect of intervention frequency on FMS

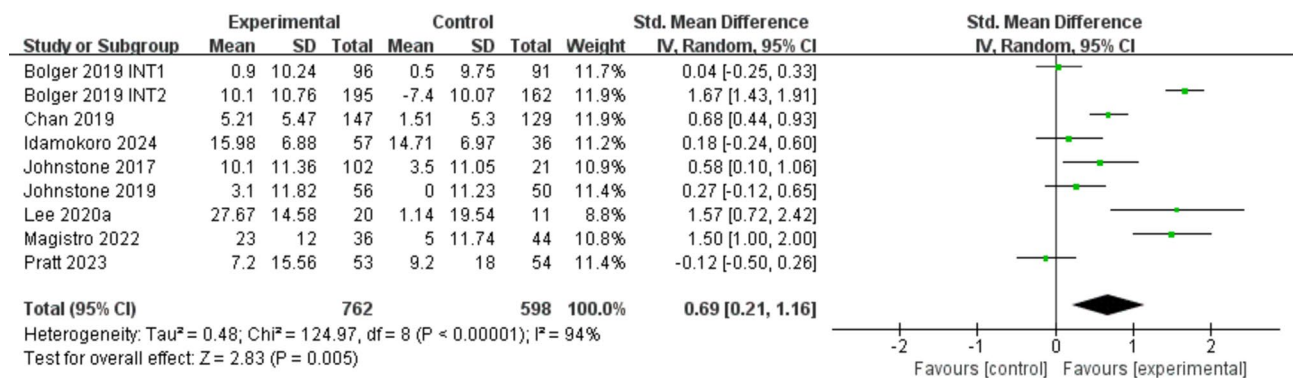
As shown in Fig. 3b, according to the results of the studies included in the meta-analysis, the intervention frequency of once a week (SMD = 0.36, 95% CI = -0.02–0.74,  $I^2 = 77\%$ ) did not indicate a significant effect on FMS. The intervention frequency of twice a week (SMD = 0.64, 95% CI = -0.51–1.78,  $I^2 = 98\%$ ) did not indicate a significant effect on FMS. The intervention frequency of 3 times a week or more (SMD = 1.52, 95% CI = 1.09–1.95,  $I^2 = 0\%$ ) indicated a significant effect on FMS.

### Effect of intervention period on FMS

Figure 3c indicated that the intervention period of 6–10 weeks (SMD = 0.44, 95% CI = -0.29–1.18,  $I^2 = 84\%$ ) on FMS was not significant. The effect of 10–20 weeks of intervention (SMD = 0.54, 95% CI = 0.28–0.80,  $I^2 = 39\%$ ) on FMS was significant. Over 20 weeks of intervention (SMD = 1.07, 95% CI = -0.08–2.21,  $I^2 = 97\%$ ) displayed no significant effect on FMS.

### Sensitivity analysis

The meta-analysis indicates that the overall intervention is effective but exhibits significant heterogeneity. We conducted a sensitivity analysis by sequentially excluding studies, yet the  $I^2$  value did not change significantly.



**Fig. 2** Forest plot of the overall effect of interventions on FMS

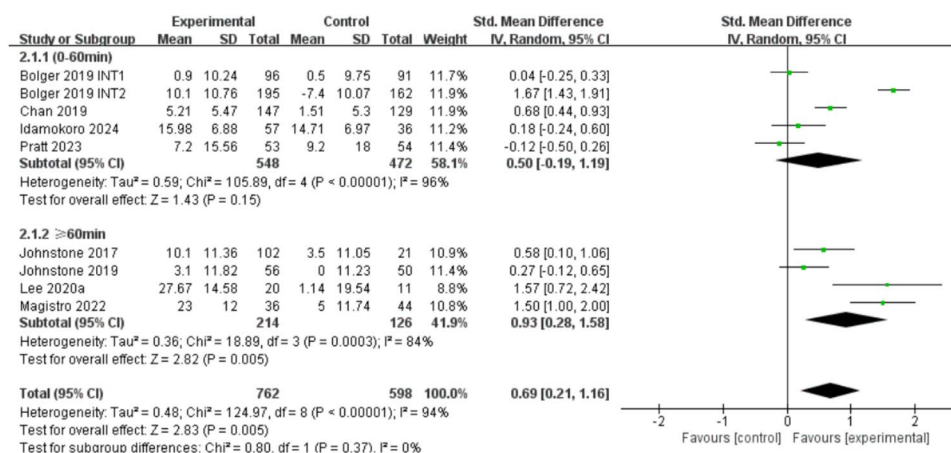


Fig.3a Forest plot of the effect of intervention duration (min) on FMS

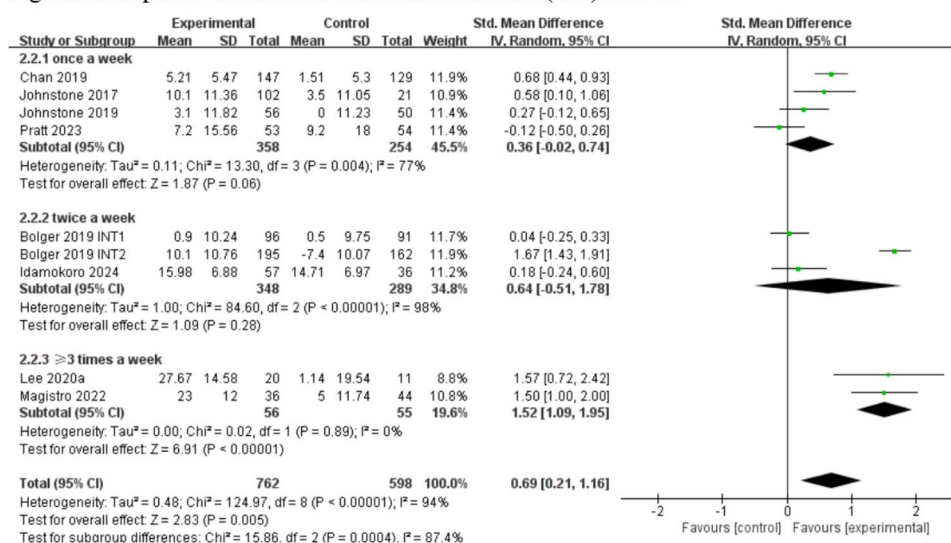


Fig.3b Forest plot of the effect of intervention frequency (time) on FMS

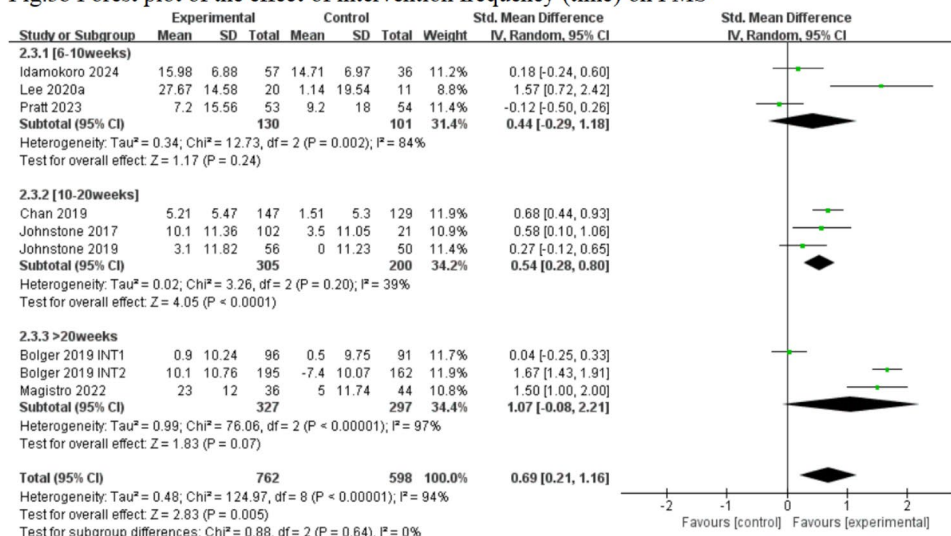


Fig.3c Forest plot of the effect of intervention period (week) on FMS

Fig. 3 Forest plot of the effect of different intervention duration, frequency, and period on FMS

**Table 3** Publication bias

| Number of studies = 9 Root MSE = 4.216 |             |           |       |       |                      |
|----------------------------------------|-------------|-----------|-------|-------|----------------------|
| Std_Eff                                | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |
| Slope                                  | 0.9364701   | 0.7654489 | 1.22  | 0.261 | −0.8735289 2.746469  |
| Bias                                   | −1.353144   | 4.342894  | −0.31 | 0.764 | −11.62246 8.916167   |

Test of H0: no small - study effects  $P=0.764$

Publication bias

As shown in Table 3, Egger’s linear regression resulted in  $|t| = 0.31$ ,  $p = 0.764$  ( $p > 0.5$ ), indicating that there is no significant publication bias in the study.

Discussion

The present study summarized the effects of school-based interventions over the last decade on enhancing FMS in children aged 6 to 12 years. Most studies present a significant effect in promoting FMS. Thirty-three studies included in this study reported on interventions for children in school settings. Findings were consistent with previous research [27] while providing new insights into the intervention period. Nevertheless, considerable variation exists in the study design, sample size, assessment tools, cultural context, and studies implemented in 14 countries across diverse settings. Interventions included in the review primarily focus on curriculum-based PA programs, which directly or indirectly influence children’s FMS.

The assessment tools for evaluating FMS in children are varied, each emphasizing different aspects of children’s skills [59]. Running, jumping, and throwing are primarily measured by the TGMD. It is simple to administer and along with established reliability and validity across multiple countries, demonstrates good applicability [60]. However, it does not encompass all three dimensions of FMS comprehensively [61]. CAMSA evaluates children’s agility and movement skills through a more dynamic approach, though the CAMSA testing process is relatively complex, and may require additional validation for applicability across diverse cultural backgrounds [62]. The MABC assesses motor coordination, balance, and object control skills, revealing some cultural variations across different regions [63]. The use of different assessment tools to evaluate FMS intervention effects may lead to discrepancies due to their unique evaluative focuses. Additionally, the age ranges for these tools differ. There may be cultural differences among these assessment tools. The scoring may also diverge among evaluators, influenced by their interpretations of the standardized criteria associated with each tool. Research indicates that the TGMD demonstrates superior reliability, validity, and applicability [59]. Given the diversity of assessment tools, further investigation into FMS evaluation instruments is warranted.

The results of the meta-analysis indicated a moderate effect of the interventions on the promoting FMS. These findings bolster confidence in practical interventions, suggesting that school-based interventions in practical applications positively impact children’s FMS. However, there is still a need to explore and optimize similar interventions to enhance their effectiveness on FMS. In the meta-analysis, the effect size of some studies was significantly larger than other studies. Studies presenting substantial effects adopted diversified and multidomain intervention strategies, such as strategies include interventions integrating math and FMS [53, 58]. While the other studies were more monotone [47, 50]. Moreover, the frequency of intervention in the studies with substantial effects is significantly higher than in studies with small effects. The studies also used different intervention duration, intervention frequency, and intervention period, which are all factors that may affect the effect of each study.

A significant effect was revealed in the present study when the intervention duration was 60 min or longer. And some other studies agreed with these results [31, 38, 56]. These findings contrast with another study, which found that most interventions lasted between 40 and 60 min and both short- and long-duration interventions were effective in enhancing FMS [24]. The school environment, including lessons, games, and peer interactions, provides various opportunities for children to develop FMS [32]. In the above settings in school, the duration of interventions may vary. Each intervention duration includes several components. In addition to the FMS-focused activities, other elements, such as warm-up and relaxation periods, may also influence the intervention’s effectiveness [44, 64, 65]. One study concluded that the duration of FMS task is worth investigating, as it may influence the developmental outcomes of FMS [23]. For example, an appropriate warm-up duration can fully stimulate neuromuscular excitability. A warm-up that is too long can lead to a gradual decrease in arousal, while a warm-up that is too short may result in insufficient arousal of the organism, both of which could reduce the effectiveness of the FMS intervention. However, most studies did not report the duration of the FMS tasks. Given that the attention span of children in 6 to 12 years is limited, unstable, and varies with age [32], and interventions may include diverse components, excessive duration and content could reduce the effectiveness of FMS tasks. What’s more, according to the OPTIMAL theory [66], the optimal duration of an intervention is affected by external factors, such as the complexity of the instructions provided by teachers and researchers. More concise intervention instructions reduce the cognitive load on children’s limited working memory capacity, making it easier to develop FMS. Therefore, exploring

the correlation between overall intervention duration, specific components, neuromuscular stimulation, and external determinants is important for understanding the significance of interventions.

The results of this meta-analysis indicated a notable effect when the intervention frequency is three times per week or more. In the practical applications, some interventions over 3 times per week or more demonstrated significant effects [22, 42, 57]. However, most interventions in this systematic review were delivered once or twice a week. These findings in the meta-analysis suggest that interventions should ideally be delivered as frequently as possible. High-frequency interventions are thought to enhance neuromuscular memory, contributing to greater FMS proficiency during the retention phase [67]. Interventions occurring three times a week or more show greater effectiveness in improving FMS [68]. Given that students are often exposed to a complex environment filled with distractions, children may experience shifting interests [69]. Therefore, to maximize the outcomes of interventions, it is feasible to increase the intervention frequency to ensure sufficient stimulation and reinforce motor memories [70]. Considering the physical and cognitive development of children, organizing more frequent interventions in school settings would be beneficial.

Meta-analysis also displayed a significant effect when the intervention period lasted between 10 and 20 weeks, which was the most common period in this review. In the practical interventions, medium and long-period interventions proved to be effective in promoting the development of children's FMS [42, 44, 71]. Moreover, the intervention effect over 20 weeks is not significant. This suggests that medium- and long-period interventions are the most practical for promoting FMS proficiency. Intervention periods that are effective in improving FMS should be medium. Children are unable to master the appropriate motor skills and transfer them to practice in a short-term period [72]. The development of FMS is a gradual process that involves neural reorganization. After several weeks of training, motor memory transitions from a motor circuit to a sensorimotor circuit [73], enabling automatic movement adjustments. This transition is key to mastering FMS. Conversely, excessively long interventions can lead to a decline in participation and reduce effectiveness [74], as prolonged periods may lead to decreased interest among children. Interest plays an important role in maintaining active participation and fostering positive self-perceptions in children. While some studies suggested that the longer intervention periods over 20 weeks offer more opportunities for children to engage in PA and can enhance effectiveness [25, 75], prolonged interventions may not always be feasible or effective in sustaining children's engagement and interest.

Furthermore, over 20 weeks interventions may be subject to the "ceiling effect" [10]. Therefore, further research should fully consider a range of perspectives to identify the most optimal intervention period for enhancing FMS development.

Despite the sensitivity and subgroup analyses conducted in this study, the sources of heterogeneity were not identified. There were differences in the studies, but their effect on the overall effect size was not significant, suggesting that the results were relatively robust. In contrast, the higher heterogeneity may be caused by factors such as large age span, different intervention contents, and cultural backgrounds. In addition, although Egger's test indicated no significant publication bias, the presence of small-sample studies in the included research may still introduce publication bias due to methodological flaws inherent in such studies. Caution is warranted when generalizing the conclusions.

### Strengths and limitations

Strengths of this review include the use of qualitative and quantitative methods to comprehensively summarize school-based interventions on FMS of children. Studies included in the meta-analysis used the same series of TGMD, improving the comparability of the results. And included pre-test and post-test data from intervention and control groups. However, certain limitations of this systematic review must be acknowledged. Firstly, only the studies were published in English and included, which may have excluded relevant research in other languages. The generalizability of the findings in this study is subject to certain limitations. Our sample was restricted to healthy children aged 6 to 12 years, which may limit the applicability of the results to children at different developmental stages and those with special healthcare needs. The interventions included in the study came from 14 countries and may have regional limitations, so there may be regional and cultural differences in the expansion of intervention in different regions. Incorporating quasi-experimental studies in meta-analysis also presents certain limitations that may influence the results of the meta-analysis. Quasi-experimental studies, which lack the random assignment of participants to intervention and control groups, may introduce selection bias and confounding variables that can affect the internal validity of the findings. Despite the sensitivity and subgroup analyses conducted in this study, the sources of heterogeneity were not identified. Additionally, the review searched only three commonly used databases, potentially omitting some studies. Moreover, the quality of some included studies was suboptimal, and the results should be interpreted with caution.

## Implications and recommendations

Regarding the assessment of children's FMS, a variety of assessment tools are available. The selection of appropriate tools should be based on the characteristics of the children, and assessments should be strictly conducted according to the standards of the chosen tools. A multi-tool composite assessment approach can be adopted to obtain more comprehensive evaluation results. This study focuses solely on healthy children. However, future research on the FMS of children with special needs is also highly necessary. Additionally, improvements in blinding methods should be emphasized. Although implementing double-blind experiments in school settings is challenging, single-blind designs can be adopted to enhance the quality of intervention studies and ensure their accuracy and reliability. We also anticipate that future research will address the limitations of meta-regression and further deepen the understanding of this topic. Future research should also clarify the specific components and timing of interventions in school settings, and determine the relationship between different intervention measures and their specific components and intervention outcomes.

## Conclusions

In summary, this systematic review and meta-analysis demonstrate that school-based interventions significantly improve FMS in children. The current studies included in the analysis show limitations in intervention characteristics: 25% of studies had interventions lasting more than 60 min, nearly 35% had a frequency of once a week, and around 50% had intervention periods of less than 10 weeks. These may be limitations in the maximum impact of interventions. The findings of the meta-analysis suggest that interventions with longer duration, higher frequencies, and medium- to long-term periods may provide a strong theoretical and practical foundation for promoting FMS in school settings. Furthermore, as critical factors, the intervention duration, frequency, and period and their combinations should be explored in greater depth in future studies.

## Abbreviations

|             |                                                                    |
|-------------|--------------------------------------------------------------------|
| AP          | Active Play                                                        |
| BOT-2       | Bruininks-Oseretsky Test of Motor Proficiency-Second Edition       |
| CAMSA       | Canadian Agility Movement Skill Assessment                         |
| CON         | Control Group                                                      |
| FMS         | Fundamental Movement Skills                                        |
| FMSQ        | Fundamental Motor Skills Quotient                                  |
| FMS-polygon | Fundamental Movement Skill Polygon                                 |
| INT         | Intervention Group                                                 |
| KTK         | Körperkoordinationstest Für Kinder                                 |
| MABC-2      | Movement Assessment Battery for Children-Second Edition            |
| MCA         | Motor Competence Assessment                                        |
| PA          | Physical Activities                                                |
| PE          | Physical Education                                                 |
| PEDro       | Physiotherapy Evidence Database                                    |
| PRISMA      | Preferred Reporting Items for Systematic Reviews and Meta-Analyses |

|          |                                                          |
|----------|----------------------------------------------------------|
| PROSPERO | International Prospective Register for Systematic Review |
| RCT      | Randomized Controlled Trial                              |
| SD       | Standard Deviation                                       |
| SE       | Standard Error                                           |
| SMD      | Standardized Mean Difference                             |
| TGMD     | Test of Gross Motor Development                          |
| TGMD-2   | Test of Gross Motor Development, Second Edition          |
| TGMD-3   | Test of Gross Motor Development, Third Edition           |
| WHO      | World Health Organization                                |
| 95% CI   | 95% Confidence Interval                                  |

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Not applicable.

## Author contributions

YL designed the study and drafted the framework of this study. XYY drafted the manuscript, extracted data and completed all figures and tables and revised figures, tables and manuscript. DQZ: designed the framework of this study and edited the manuscript. YCW: extracted data and revised figures, tables, designed the study, and checked all data and edited the manuscript. YYS: edited the manuscript. ZHW: edited the manuscript. All authors approve the final version of the manuscript submitted for publication.

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## Data availability

The data used and /or analyzed in the current study is available from the first author on reasonable request.

## Declarations

## Ethics approval and consent to participate

Not applicable.

## Consent for publication

Not applicable.

## Conflict of interest

The authors declare no conflicts of interest.

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